


Penalized Regression Methods with Applications to the General Linear Models



Topics:

- ❖ **General Linear Regression & Classical Variable Selection Methods:**
 - Stepwise variable selection
 - Best subset variable selection
- ❖ **Modern Variable Selection Methods -- Penalized Regression:**
 - Ridge Regression
 - LASSO
 - Elastic Net
- ❖ **The Latest Verdict & The Link to How to do it in R!**

Classical Linear Regression & Variable Selection Methods



General Linear Models

A typical general linear model with response variable Y and predictors $X_1, \dots, X_p$ is written as:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \dots + \beta_p X_{ip} + \epsilon_i$$

where β are regression parameters, ϵ is the random noise term, and i is the observation number, $i = 1, \dots, N$

The Ordinary Least Squares estimators correspond to the values for the coefficients (β) that will minimize the Residual Sum of Squares (RSS):

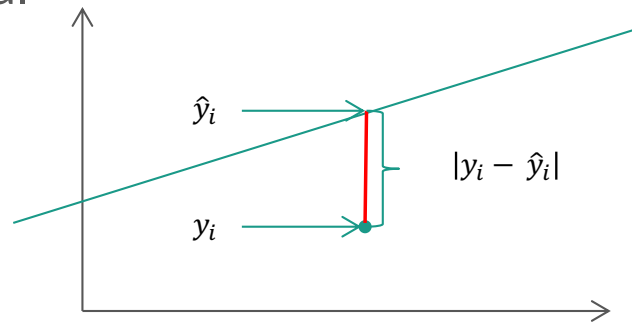
$$\text{RSS}(\beta) = \sum_{i=1}^N (Y_i - \beta_0 - \beta_1 X_{i1} - \dots - \beta_p X_{ip})^2$$

- General Linear models in matrix & vector format:

$$Y_{(N \times 1)} = \mathbf{X}_{(N \times P)} \beta_{(P \times 1)} + \epsilon_{(N \times 1)}$$

- Parameter estimation: estimate β
- Subset selection: find $< P$ variables to include in the model

- Parameter estimation: estimate β
- Residual



- The residual sum of squares (RSS):

$$\text{RSS}(\beta) = \sum_{i=1}^N (y_i - \hat{y}_i)^2 = \|y - X\beta\|^2$$

$$\hat{\beta}^{OLS} = \arg \min_{\beta} \|y - X\beta\|^2 = (X'X)^{-1}X'y$$



Popular Classical Variable Selection Methods

Stepwise variable selection:

- Forward selection and Backward selection
- First identify a subset of predictor variables by successively adding or removing variables (or both), and then fit a model on the reduced set of variables
- Computationally efficient fitting a total of $1 + p(p + 1)/2$ models
- Fast but not optimal

Best subset variable selection:

- Fit the best 1-subset, 2-subset, ... k-subset, ..., until up to all p variables
- Choose the smallest optimal (best) subset
- Computationally expensive fitting a total of 2^p models
- Optimal but slow

Modern Variable Selection Methods --

Penalized Regression Methods:

- Ridge Regression
- LASSO
- Adaptive Lasso (*optional, not in the exams*)
- Elastic Net



Modern Variable Selection Methods

- ❖ **Penalized Regression (Regularization, Shrinkage):** (1) Ridge Regression, (2) LASSO, (*optional*) adaptive LASSO, and (3) Elastic Net
- ❖ Minimize the RSS by using a penalty on the magnitudes of the regression coefficients
- ❖ Good prediction accuracy and computational efficiency
- ❖ often preferred to traditional selection methods



Bias-Variance Tradeoff

For a model M with regression coefficients $\tilde{\beta}$, the model's MSE is equal to the sum of its variance and bias as shown by the equation:

$$\begin{aligned}\text{MSE}(M) &= E \left(Y_{\text{new}} - (\tilde{\beta}_0 + \sum_{i=1}^{p-1} \tilde{\beta}_i X_{\text{new},i}) \right)^2 \\ &= \text{Var} \left(Y_{\text{new}} - (\tilde{\beta}_0 + \sum_{i=1}^{p-1} \tilde{\beta}_i X_{\text{new},i}) \right) + \text{Bias}(\tilde{\beta})^2\end{aligned}$$

where $(X_{\text{new}}, Y_{\text{new}})$ represents new data that are not used to obtain the coefficient estimates $\tilde{\beta}$.



Bias-Variance Tradeoff

$$\begin{aligned}\text{MSE}(\mathbf{M}) &= E \left(Y_{\text{new}} - (\tilde{\beta}_0 + \sum_{j=1}^{p-1} \tilde{\beta}_j X_{\text{new},j}) \right)^2 \\ &= \text{Var} \left(Y_{\text{new}} - (\tilde{\beta}_0 + \sum_{j=1}^{p-1} \tilde{\beta}_j X_{\text{new},j}) \right) + \text{Bias}(\tilde{\beta})^2\end{aligned}$$

Bias-Variance Trade-off: For penalized regression methods: bias $\uparrow$, but variance $\downarrow$.

Often, the increase in bias is less than the decrease in variance; hence, the resulting model would have a smaller MSE than the unbiased OLS model. Therefore, penalized regression methods can produce models that have stronger predictive performance for the new data.



Penalized Regression Methods

- Penalized regression methods keep all the predictor variables in the model but constrain the regression coefficients by shrinking them toward zero.
- If the amount of shrinkage is large enough, these methods can perform **variable selection** by shrinking some coefficients to 0.
- The general form of the shrinkage and regularization methods for linear models:

$$\tilde{\beta} = \arg \min_{\beta} \left(\sum_{i=1}^N (y_i - (X\beta)_i)^2 + \lambda P(\beta) \right)$$

where $\lambda \geq 0$ is shrinkage parameter, $P(\beta)$ is a penalty on the regression coefficient.

λ controls the size of the coefficients

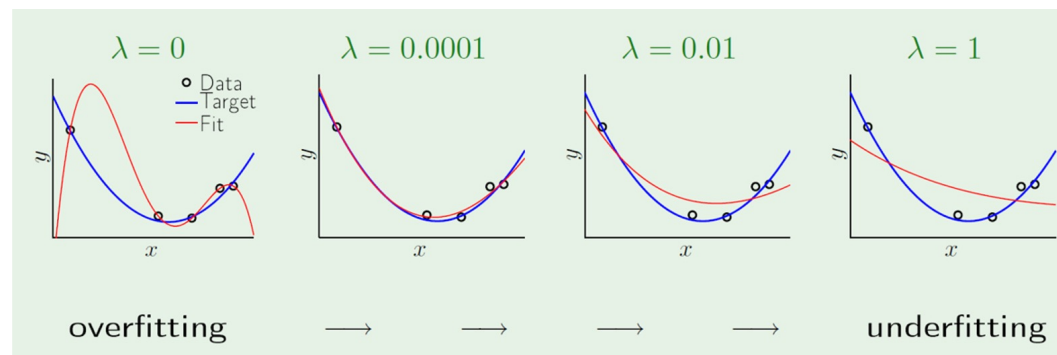
λ controls amount of regularization

As $\lambda \downarrow 0$, we obtain the least squares solutions

As $\lambda \uparrow \infty$, we have $\beta = 0$ (intercept-only model)

Penalized Regression Methods

- Penalized regression methods are also known as shrinkage or regularization methods.
- Regularization is our first weapon to combat overfitting.
- Look at what a little regularization can do (Nathan Bastian, Northwestern University):





Penalized Regression Methods

A penalized regression method produces a series of models, $M_0, M_1, \dots, M_k$, where each model is the solution for a unique tuning parameter value:

- M_0 : the least complex model, the maximum amount of penalty is imposed.
- M_k : the most complex model, for which no penalty is imposed. Which means?



Penalized Regression Methods

Choose a model: the one yields the minimum prediction error.

- **Prediction error** of a model can be estimated **directly** or **indirectly**.
- **Direct techniques** estimate the prediction error by scoring validation data or by using cross validation.
- **Indirect methods** estimate the prediction error by making mathematical adjustments to the RSS.
 - AIC: (Akaike information criterion)
 - BIC: (Bayesian information criterion)
 - Mallow's C_p (equivalent to AIC for linear regression)

Review of Penalized Regression, in the matrix/vector form

- Reasons for variable selection:
 - Principle of parsimony
 - Interpretation: each β_j should be meaningful
 - Prediction: large P leads to overfit.

- Shrinkage methods

$$\hat{\beta}(\lambda) = \arg \min_{\beta} \{\|y - X\beta\|^2 + \lambda J(\beta)\}$$

- Equivalently

$$\hat{\beta}(t) = \arg \min_{\beta} \|y - X\beta\|^2 \text{ subject to } J(\beta) \leq t$$

Review, Penalty terms in the matrix/vector form

- Ridge

$$J(\beta) = \|\beta\|^2$$

- LASSO

$$J(\beta) = \|\beta\|_1$$

- Elastic net

$$\lambda J(\beta) = \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|^2$$

Ridge Regression

—



Ridge Regression

If the β_j 's are unconstrained:

- They can explode
- Tend to have high variance

To control variance, we can regularize the coefficients:

Might impose the ridge constraint:

Minimize: $\sum_1^n (y_i - (X\beta)_i)^2$ s.t. $\sum_1^p \beta_j^2 \leq t$ (t is the shrinkage parameter)

By convention (very important!):

- X is assumed to be standardized (mean 0, unit variance) **why?**
- y is assumed to be centered



Ridge Regression

Because the ridge regression coefficient estimates can change *substantially* when multiplying a given predictor by a constant, due to the sum of squared coefficients term in the penalty part of the ridge regression objective function.

Ridge Regression

We can write the ridge constraint as the following penalized residual sum of squares (PRSS):

Compound optimization format:

$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p \beta_j^2 = \text{RSS} + \lambda \sum_{j=1}^p \beta_j^2$$

Constrained optimization format:

$$\hat{\beta}^{\text{ridge}} = \underset{\beta}{\operatorname{argmin}} \sum_{i=1}^N \left(y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j \right)^2$$

subject to $\sum_{j=1}^p \beta_j^2 \leq t,$

where $\lambda \geq 0$ is tuning parameter:

- λ controls the size of the coefficients

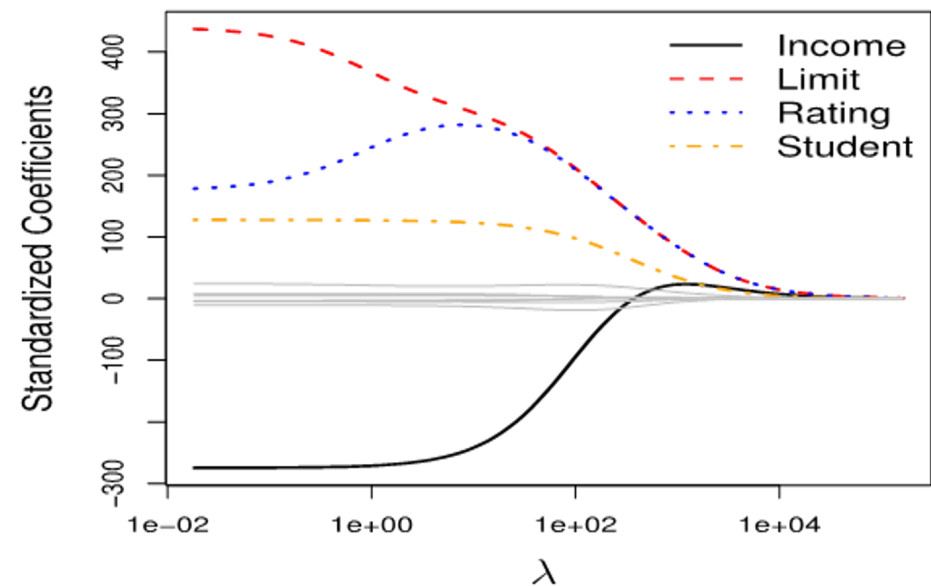
- λ controls amount of regularization

- As $\lambda \downarrow 0$, we obtain the least squares solutions

- As $\lambda \uparrow \infty$, we have $\hat{\beta}^{\text{ridge}}_{\lambda=\infty} = 0$ (intercept-only model)

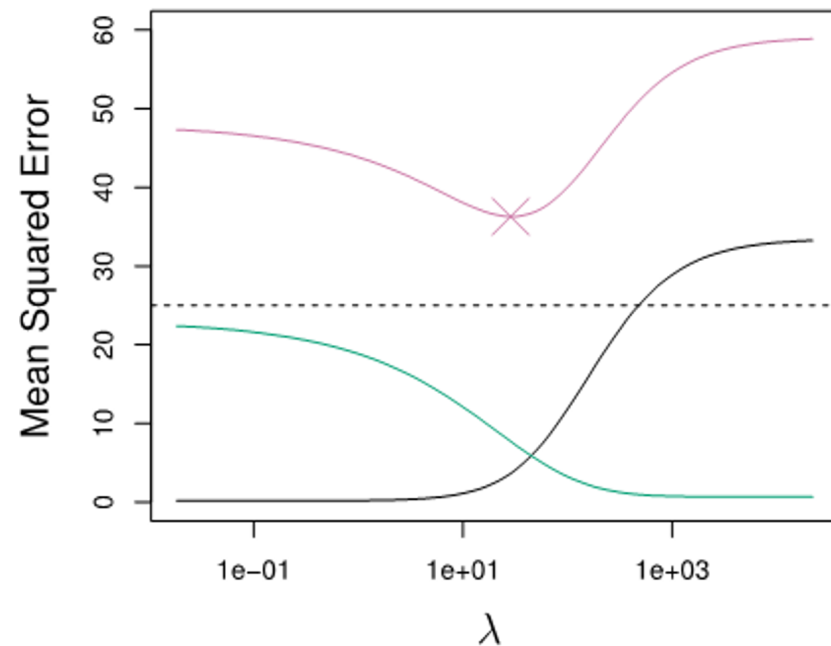
Ridge Regression

- As λ increases, the standardized ridge regression coefficients shrink towards zero (Nathan Bastian, Northwestern University).
- Thus, when λ is extremely large, then all of the ridge coefficient estimates are basically zero; this corresponds to the *null model* that contains no predictors.



Ridge Regression

- Black = Bias
- Green = Variance
- Purple = MSE
- ★ Increased λ leads to increased bias but decreased variance.
- ★ Ridge regression will work best in situations where the OLS estimates have high variance (Nathan Bastian, Northwestern University).



Review: Ridge Regression, in matrix & vector form

- $\hat{\beta}^{ridge} = \arg \min_{\beta} \{\|y - X\beta\|^2 + \lambda \|\beta\|^2\}$
- Equivalently
- $\hat{\beta}^{ridge} = \arg \min_{\beta} \|y - X\beta\|^2$ subject to $\|\beta\|^2 \leq t$
- Pros: a stable regression in the underdetermined system
- Cons: using all the variables

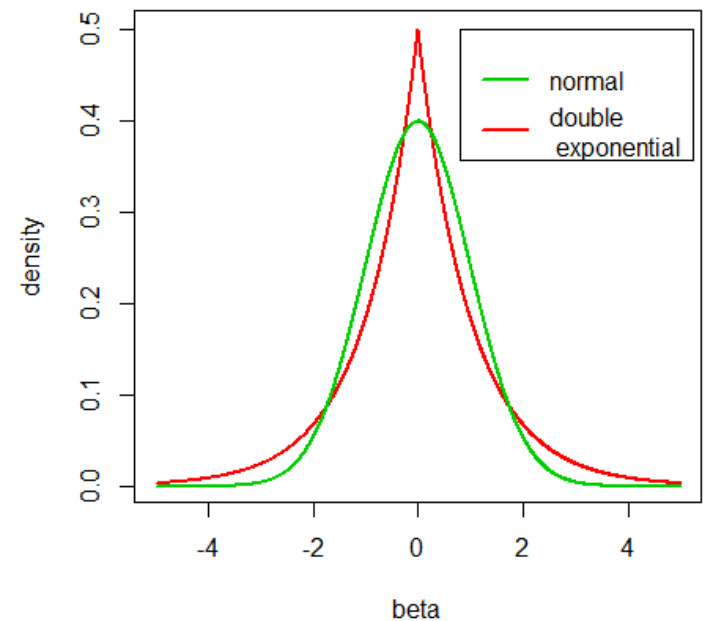
Review: Ridge Regression

❖ Explicit OLS Solution:

$$\hat{\beta}^{ridge} = (X'X + \lambda I)^{-1}X'y$$

❖ Bayesian Interpretation:

$\hat{\beta}^{ridge}$ can also be interpreted as the posterior estimator when the prior on β follows the normal distribution



LASSO

--- Least **Absolute Shrinkage** and **Selection Operator**



Why use LASSO?

- LASSO stands for Least Absolute Shrinkage and Selection Operator
- The penalty term in ridge regression will never force the coefficients to be exactly zero. As a result, the final model includes every p predictor making model interpretation difficult
- LASSO works similarly to ridge regression but uses a different penalty term so **some coefficients can go to zero**



Goal of LASSO is to minimize the coefficients

$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p |\beta_j| = \text{RSS} + \lambda \sum_{j=1}^p |\beta_j|$$

- Where RSS is residual sum of squares and λ is the tuning parameter which controls the amount of regularization
- Lasso uses a L1 penalty instead of a L2 penalty that ridge regression uses, which forces some of the coefficients to be exactly zero when λ gets very large



LASSO and Ridge Regression Coefficient Estimates

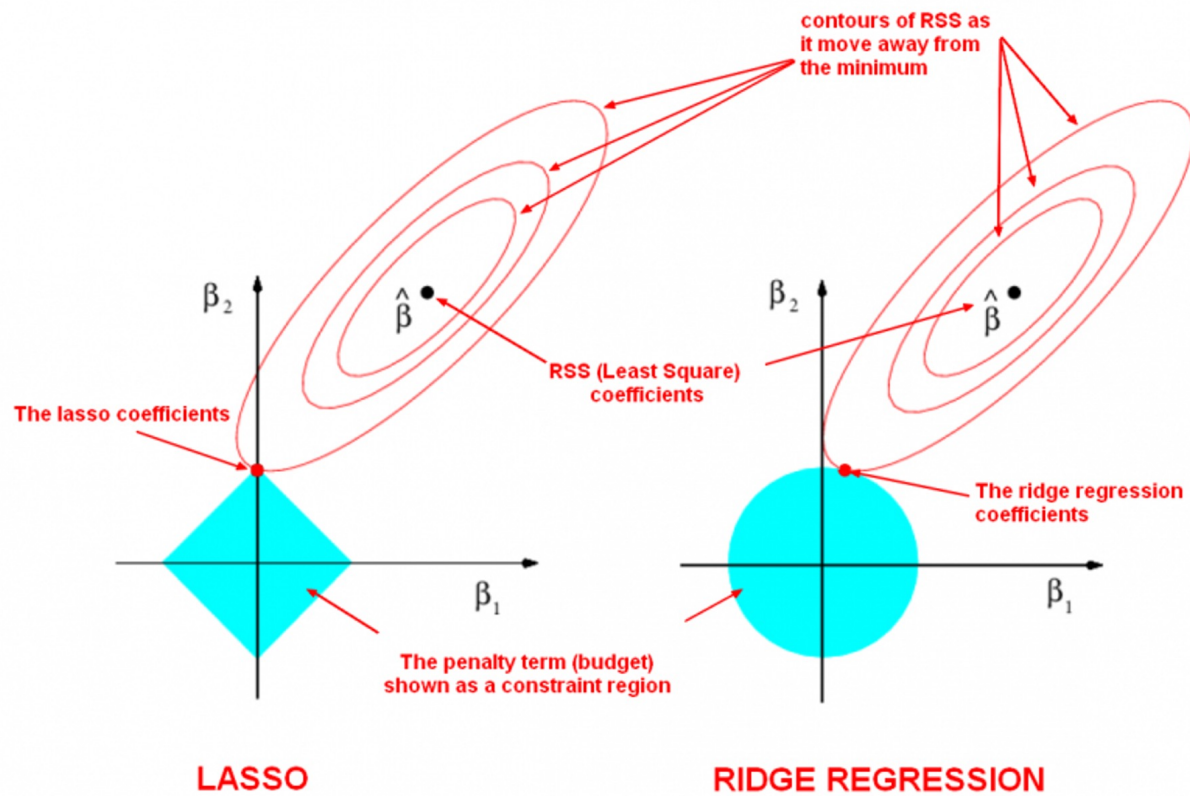
The coefficient estimates of LASSO and Ridge solve

$$\underset{\beta}{\text{minimize}} \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 \quad \text{subject to} \quad \sum_{j=1}^p |\beta_j| \leq s$$

and

$$\underset{\beta}{\text{minimize}} \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 \quad \text{subject to} \quad \sum_{j=1}^p \beta_j^2 \leq s,$$

Visualizing the solutions



How can we choose the tuning parameter λ ?



In subset selection for Ridge Regression and LASSO we need to find which of the models we are considering is best so we need some way of picking a value for λ which is equivalent to picking a value for constraint s .

Steps to select λ or s

1. Select a grid of potential values
2. Use cross-validation to estimate error rate on test data (for each value of λ)
3. Select the value that gives the smallest error rate
4. Re-fit the model using all the variable observations and selected value of λ

Simulated Data Set SAS Code

```
data Simdata;  
  drop i j;  
  array x{5} x1-x5;  
  do i=1 to 1000;  
    do j=1 to 5;  
      x{j} = ranuni(1); /* Continuous predictors */  
    end;  
    c1 = int(1.5+ranuni(1)*7); /* Classification variables */  
    c2 = 1 + mod(i,3);  
    yTrue = 2 + 5*x2 - 17*x1*x2 + 6*(c1=2) + 5*(c1=5);  
    y = yTrue + 2*rannor(1);  
    output Simdata;  
  end;  
  
run;
```

Simulated Data Set SAS Code (cont.)

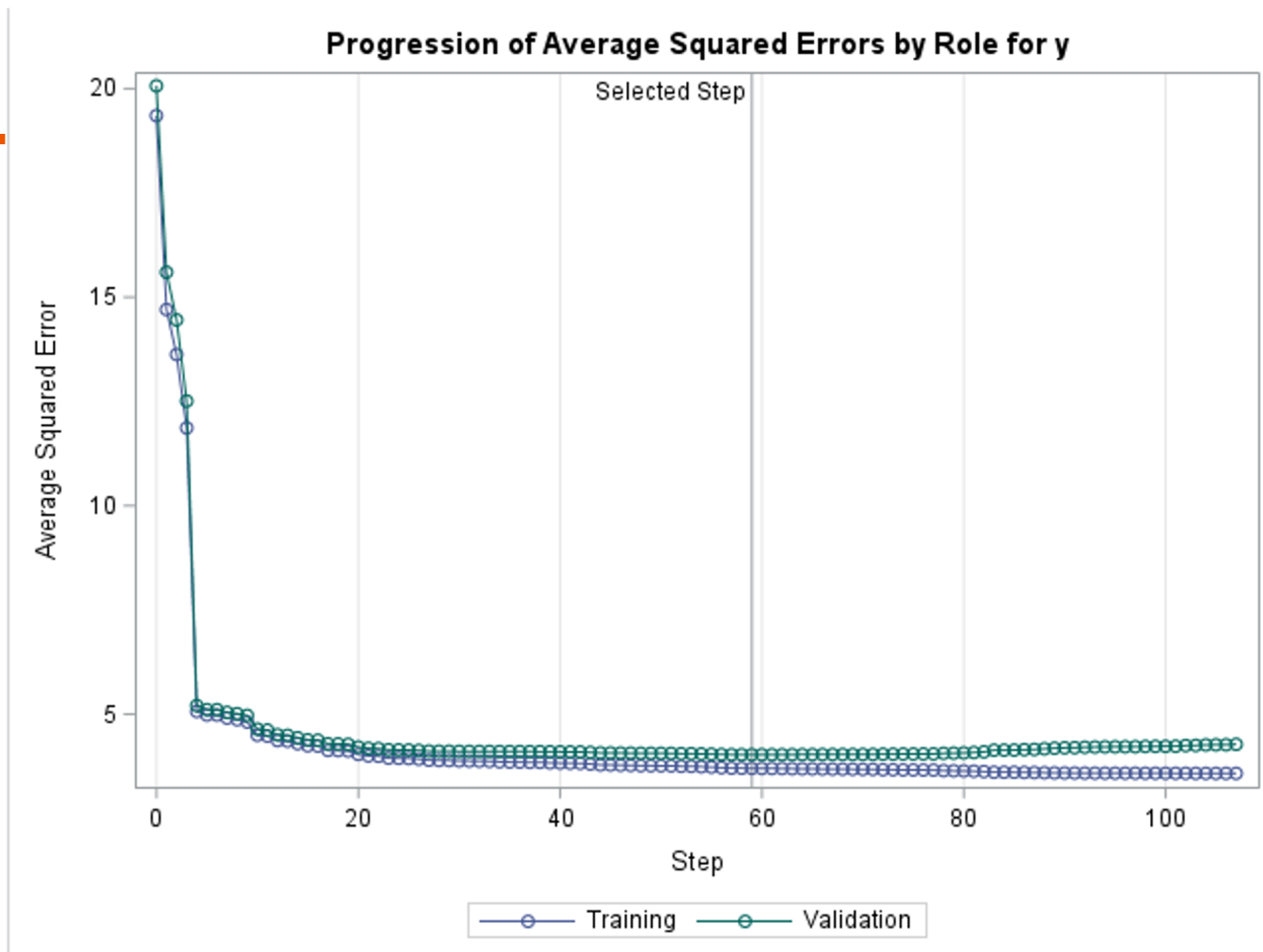
- Simdata contains 1000 observations, five continuous variables (x1,...,x5) and two classification variables (c1 and c2). The response variable is generated from a linear function of only x1, x2, and c1.
- The following GLMSELECT procedure performs a LASSO fit for the data. The validation data approach is used as a tuning method

```
proc glmselect data=Simdata plots=all;  
  partition fraction(validate=.3);  
  class c1 c2;  
  model y = c1|c2|x1|x2|x3|x4|x5 @2  
          / selection=lasso(stop=none choose=validate);  
run;
```

Interpreting the code



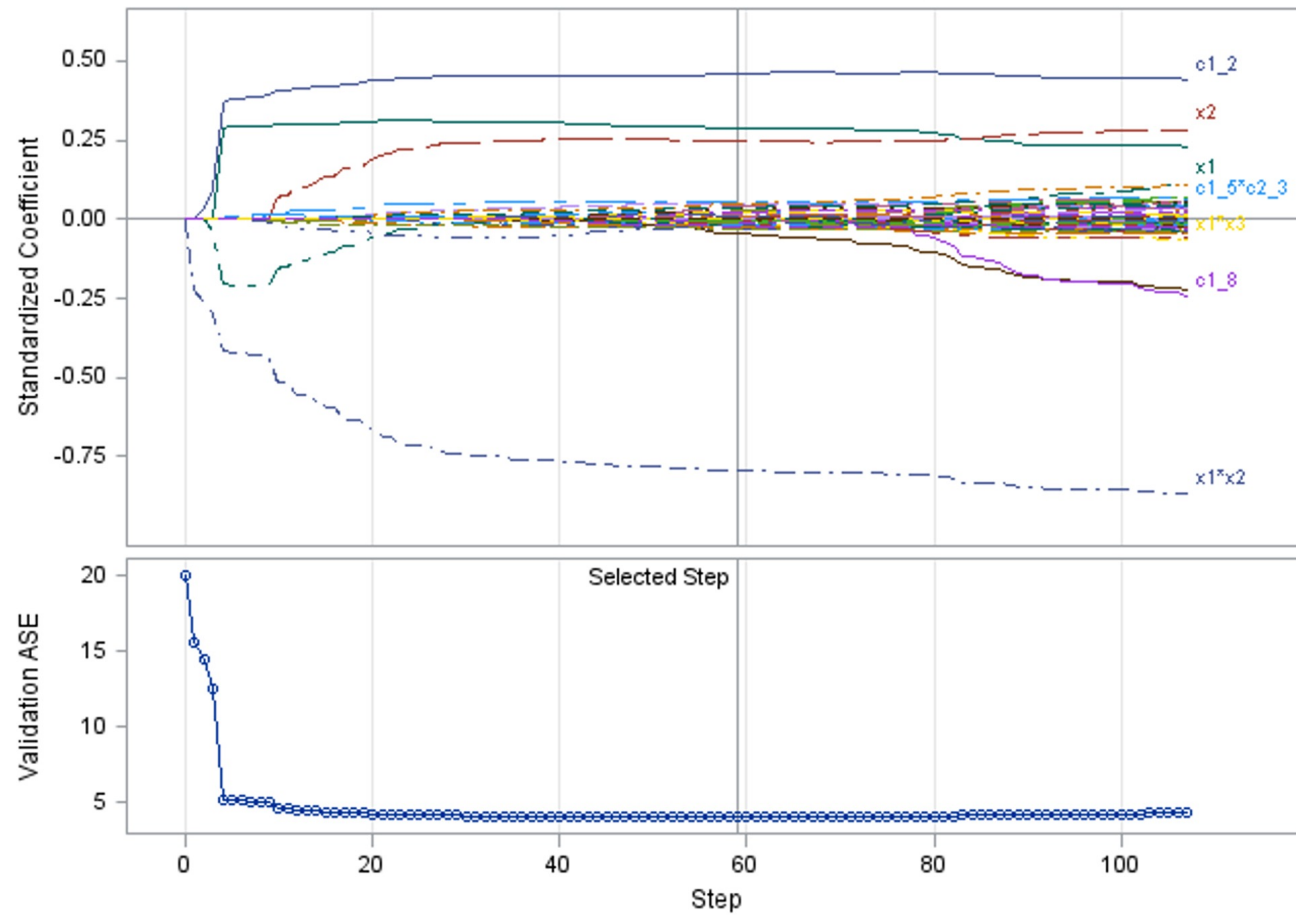
The MODEL statement request that a linear model be built using all the effects (c1, c2, x1, x2, x3, x4 and x5) and their two-way interactions. The PARTITION statement randomly reserves 30% of the data as validation data and uses the remaining 70% as training data. The training set is used for fitting the models, and the validation set is used for estimating the prediction error for model selection. The SELECTION=LASSO option in the MODEL statement requests LASSO selection. The CHOOSE=VALIDATE suboption in the MODEL statement requests that validation data be used as the tuning method for the LASSO selection. If you have enough data, using a validation data set is the best way to tune a penalized regression method. The observations in the training set are used to produce a LASSO solution path ($M_0, \dots, M_k$), and the observations in the validation set are used to estimate the prediction error of each model (M_i) on the solution path. The model that yields the smallest ASE on the validation data is then selected.



Graph explanation

- Shown above is the ASE (Average Square Estimates) of the models on the LASSO solution path separately for the training and validation sets. The LASSO solution path is created on a grid of the tuning parameter value (t). The X axis shows the normalized tuning values, which are the L1 norm of the regression coefficients divided by the L1 norm of the ordinary least squares solution.
- As you move from left to right on the X axis, the amount of shrinkage that is imposed on the regression coefficients decreases. Hence, the model complexity increases from the null model to the full OLS model, with all 106 regression parameters. As the model complexity increases, the ASE on the training data consistently decreases, typically dropping to 0 if you increase the model complexity enough. However, a model that has a training error of 0 is overfit to the training data and will typically generalize poorly on a new data set. Thus, training error is not a good estimate of the prediction error. On the other hand, the prediction error on the validation data first decreases, but then increases; the point of minimum ASE is marked by the vertical line on the plot, at a tuning value of about 0.44. The decrease in the validation error indicates that the effects that join the model before the vertical line are probably the important effects in explaining the variation in the response variable. The subsequent increase indicates that the later effects describe the random noise in the training data, which is not reproducible in the validation

Coefficient Progression for y



Graph Explanation



- Shown above is the standardized coefficient estimates as a function of tuning parameter values. It shows that the first variable that joins the model is the interaction effect between x_1 and x_2 (shown by the dashed yellow line), the second variable is level 2 of the classification variable c_1 (shown by the solid red line), and so on. The Y axis shows the standardized coefficient estimates. The vertical line above once again indicates the LASSO fit for which the prediction error on the validation data is the smallest.
- The SAS output (which isn't copied here) shows that the selected model includes 33 effects when the normalized tuning value is 0.5. However, on close inspection the validation error in the graph above does not change much after 0.4.

Review: LASSO in matrix & vector form

Compound optimization format:

$$\hat{\beta}^{LASSO} = \arg \min_{\beta} \{ \|y - X\beta\|^2 + \lambda \|\beta\|_1 \}$$

- Equivalently

Constrained optimization format:

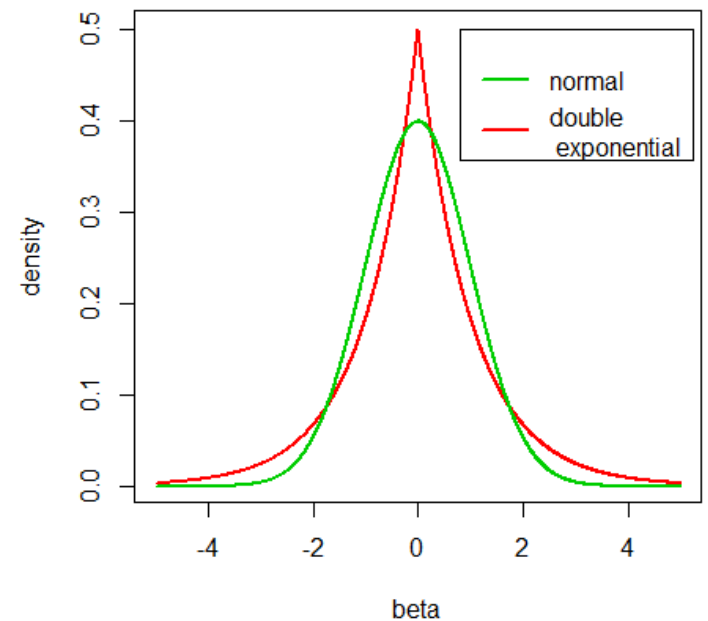
$$\hat{\beta}^{LASSO} = \arg \min_{\beta} \|y - X\beta\|^2 \text{ subject to } \|\beta\|_1 \leq t$$

- $\|\beta\|_1 = \sum_{j=1}^P |\beta_j|$ - L_1 norm
- No closed form expression is available in general.
- Numerically, this is a convex optimization problem, which could be solved via software.

Review: LASSO

❖ Bayesian Interpretation:

$\hat{\beta}^{LASSO}$ can also be interpreted as the posterior estimator when the prior on β follows the Laplace distribution (also referred to as the double exponential distribution)



Review: LASSO

- ❖ Pros: LASSO performs parameter estimation and variable selection simultaneously
- ❖ Cons:
 - When the correlations within a group of predictors are very high, LASSO tends to select only one variable from the group
 - Estimation consistency and selection consistency usually cannot be obtained simultaneously
 - Some theoretical conditions apply only in the extreme $P \gg N$
 - LASSO is biased

Elastic Net

1. Introduction and Simple Simulation Example
2. Cross Validation and External Cross Validation
3. The turning Values
4. Prostate Data Example





Introduction and Simple Simulation Example

Although LASSO selection performs well for a wide range of variable selection problems, it has some limitations.

- The number of predictor variables it selects cannot exceed the sample size.
- LASSO occurs when there are groups of correlated variables. It fails to do a group selection by selecting only one variable from a group and ignoring the others.

Elastic net removes the limitation on the number of selected variables and performs group selection by selecting all the variables that form the group (Zou and Hastie 2005).



Elastic net solves the following optimization problem:

Constrained optimization format:

$$\tilde{\beta} = \arg \min_{\beta} \sum_{i=1}^N (y_i - (X\beta)_i)^2 \quad \text{subject to} \quad \sum_{j=1}^p |\beta_j| \leq t_1 \text{ and } \sum_{j=1}^p \beta_j^2 \leq t_2$$

where the penalty is placed on both the L_1 norm ($\sum_{j=1}^p |\beta_j|$) and the L_2 norm ($\sum_{j=1}^p \beta_j^2$) of the regression coefficients. The L_1 part of the penalty performs variable selection by setting some coefficients to exactly 0, and the L_2 part of the penalty encourages the group selection by shrinking the coefficients of correlated variables toward each other.

The same problem can be rewritten in the following Lagrangian form, where λ_1 and λ_2 are the tuning parameters:

$$\tilde{\beta} = \arg \min_{\beta} \left(\sum_{i=1}^N (y_i - (X\beta)_i)^2 + \lambda_1 \sum_{j=1}^p |\beta_j| + \lambda_2 \sum_{j=1}^p \beta_j^2 \right) \quad \Leftarrow \text{Compound optimization format}$$



Simple Simulation Example

$$z_1, z_2 \sim \text{uniform}(0, 20)$$

The response vector is generated by:

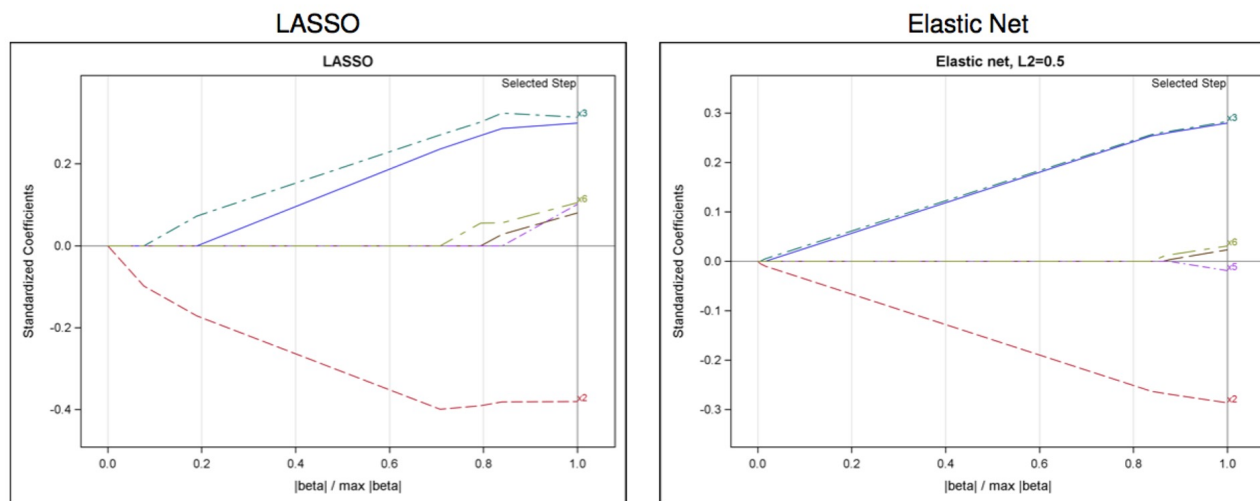
$$y = z_1 + 0.1z_2 + N(0, 1)$$

Suppose the observed predictors ($x_1, x_2, \dots, x_6$) are generated from the “hidden” factors (z_1, z_2) in the following way:

$$x_1 = z_1 + \epsilon_1, \quad x_2 = -z_1 + \epsilon_2, \quad x_3 = z_1 + \epsilon_3$$

$$x_4 = z_2 + \epsilon_4, \quad x_5 = -z_2 + \epsilon_5, \quad x_6 = z_2 + \epsilon_6$$

Based on the simulation setup, for a linear model where y is the response variable and $x_1, \dots, x_6$ are the explanatory variables, a good selection procedure would identify x_1, x_2 , and x_3 (the z_1 group) together as the most important variables. The figure shows coefficient progression plots that are generated by LASSO and elastic net selection.





Cross Validation and External Cross Validation

Cross validation uses part of the training data to fit the model and a different part to estimate the prediction error.

- For a k-fold cross validation, you split the data into k approximately equal-sized disjoint parts. You reserve one part of the data for validation, and you fit the model to the remaining k-1 parts of the data. Then you use this fitted model to calculate the prediction error for the reserved part of the data.
- You do this for all k parts, and summation of k estimates of the prediction error divided by the total training sample size yields the k-fold cross validation error.



CV and CVEX

- The regular cross validation method (CV) that is available in the GLMSELECT procedure uses the OLS estimation to obtain the prediction error for each of the $k-1$ parts of the data, regardless of which estimation method is specified in the MODEL statement.
- The external cross validation method (CVEX) uses the selection method specified in the MODEL statement to estimate the cross validation error.

Therefore, for penalized regression problems, you should specify CVEX instead of CV in the CHOOSE= option.



Choosing the Tuning Values for Elastic Net

- In elastic net, the penalty is placed on both the L1 norm and the L2 norm of the regression coefficients; hence you must specify methods to choose both tuning parameters, λ_1 and λ_2 .
- A typical approach for choosing the tuning values is to first run the analysis on a grid of λ_2 values, such as (0, 0.01, 0.1, 1, 10, and 100). For each λ_2 value, the GLMSELECT procedure uses the efficient LARS-EN algorithm to produce the whole solution path that depend on λ_1 . Then for each λ_2 value, you can choose the λ_1 value that yields the smallest k-fold cross validation error. After this step, you can choose the λ_2 value that yields the minimum k-fold cross validation error.

Prostate Data

Figure 9 Correlation Matrix for the Prostate Data

Pearson Correlation Coefficients, N = 97 Prob > r under H0: Rho=0								
	lcavol	lweight	age	lbph	svi	lcp	gleason	pgg45
lcavol	1.00000	0.28052 0.0054	0.22500 0.0267	0.02735 0.7903	0.53885 <.0001	0.67531 <.0001	0.43242 <.0001	0.43365 <.0001
lweight	0.28052 0.0054	1.00000	0.34797 0.0005	0.44226 <.0001	0.15538 0.1286	0.16454 0.1073	0.05688 0.5800	0.10735 0.2953
age	0.22500 0.0267	0.34797 0.0005	1.00000	0.35019 0.0004	0.11766 0.2511	0.12767 0.2127	0.26889 0.0077	0.27611 0.0062
lbph	0.02735 0.7903	0.44226 <.0001	0.35019 0.0004	1.00000	-0.08584 0.4031	-0.00700 0.9458	0.07782 0.4487	0.07846 0.4449
svi	0.53885 <.0001	0.15538 0.1286	0.11766 0.2511	-0.08584 0.4031	1.00000	0.67311 <.0001	0.32041 0.0014	0.45765 <.0001
lcp	0.67531 <.0001	0.16454 0.1073	0.12767 0.2127	-0.00700 0.9458	0.67311 <.0001	1.00000	0.51483 <.0001	0.63153 <.0001
gleason	0.43242 <.0001	0.05688 0.5800	0.26889 0.0077	0.07782 0.4487	0.32041 0.0014	0.51483 <.0001	1.00000	0.75190 <.0001
pgg45	0.43365 <.0001	0.10735 0.2953	0.27611 0.0062	0.07846 0.4449	0.45765 <.0001	0.63153 <.0001	0.75190 <.0001	1.00000

The figure 9 displays the correlation matrix for the eight predictors of the Prostate data. We can see that there is some significant correlation between the predictor variables, where the highest is 0.75 (between gleason and pgg45). Because of this correlation between the predictor variables, elastic net selection might be more suitable for analyzing the prostate data.



```
proc glmselect data=TrainData testdata=TestData plots(stepaxis=normb)=all;  
model lpsa = lcavol lweight age lbph svi lcp gleason pgg45  
/selection=elasticnet(L2=0.1 stop=none choose=cvex) cvmethod=split(10);  
run;
```

The L2= suboption of the SELECTION= option in the MODEL statement specifies the tuning value of 2 as 0.1, and the CHOOSE= suboption requests that external cross validation be used for determining the tuning value 1. The CVMETHOD= requests that tenfold cross validation be used.



Root MSE	0.70137
Dependent Mean	2.58256
R-Square	0.6629
Adj R-Sq	0.6476
AIC	26.22033
AICC	27.15783
SBC	-36.78569
ASE (Train)	0.46381
ASE (Test)	0.59053

Parameter Estimates		
Parameter	DF	Estimate
Intercept	1	0.044650
lcavol	1	0.497666
lweight	1	0.472119
svi	1	0.558134

Root MSE	0.68810
Dependent Mean	2.58256
R-Square	0.6755
Adj R-Sq	0.6608
AIC	23.54568
AICC	24.48318
SBC	-39.46034
ASE (Train)	0.44642
ASE (Test)	0.58521

Parameter Estimates		
Parameter	DF	Estimate
Intercept	1	-0.523866
lcavol	1	0.537463
lweight	1	0.605578
svi	1	0.670596

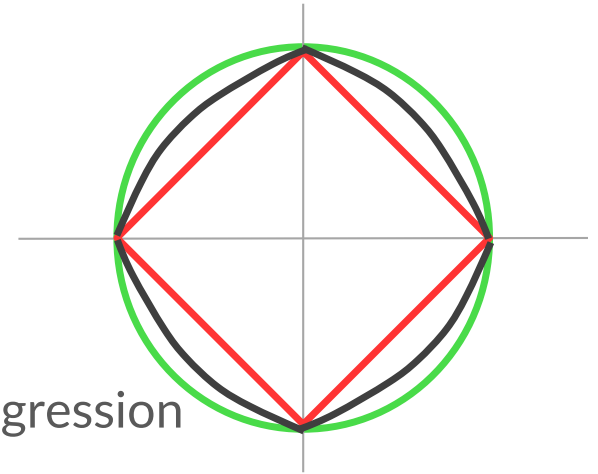
Parameter Estimates		
Parameter	DF	Estimate
Intercept	1	-0.218325
lcavol	1	0.444499
lweight	1	0.551577
svi	1	0.579602
lcp	1	0.030312
pgg45	1	0.001500

Root MSE	0.70663
Dependent Mean	2.58256
R-Square	0.6682
Adj R-Sq	0.6423
AIC	29.11264
AICC	30.91909
SBC	-29.39639
ASE (Train)	0.45653
ASE (Test)	0.58317
CVEX PRESS	0.52444

The figures from left to right are the parameter estimates and the fit statistics of the model that is selected by LASSO, Adaptive LASSO and elastic net. In addition to the three predictors (lcavol, lweight, and svi) selected by both LASSO and adaptive LASSO, elastic net chooses two more variables (lcp and pgg45). However, notice that the coefficient estimates of these two parameters are very close to 0 (0.0303 and 0.0015). The ASE of the test data is 0.58317, which is slightly less than the ASE of the models that are selected by LASSO (0.59053) and adaptive LASSO (0.58521).

Review: Elastic Net in matrix & vector form

❖ $\hat{\beta}^{elastic} = \arg \min_{\beta} \{ \|y - X\beta\|^2 + \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|^2 \}$



- ❖ Elastic net can be viewed as a combination of ridge regression and LASSO
- Select variables like the LASSO
 - Shrink the coefficients of correlated predictors together like Ridge regression